



SKAT Industrial Equipment Plant LLC (SKAT Plant LLC)

DIGITAL DIELECTRIC TEST SYSTEMS

SKAT-70C / SKAT-100C Series

AC/DC High-Voltage Insulation Test Sets

USER MANUAL

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Applies to all modifications: SKAT-70C, SKAT-70CA, SKAT-70C-K, SKAT-70CA-K, SKAT-70C-S, SKAT-70CA-S, SKAT-70C-H, SKAT-70CA-H, SKAT-100C, SKAT-100CA, SKAT-100C-S, SKAT-100CA-S

SKAT Industrial Equipment Plant LLC · 37 Poddubnogo St., Volgograd 400040, Russian Federation
Tel. +7 (8442) 26-99-94 · st@skat-v.com · www.skat-v.com

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1 Introduction

This User Manual describes the operating principle, design and construction of the SKAT-70C / SKAT-100C series digital dielectric test systems, and is intended to support their correct and safe operation.

NOTICE

Read before use

Before performing any high-voltage test with this instrument, read this User Manual in its entirety.

2 Intended Use

The SKAT series digital dielectric test systems (hereafter, "the instrument") generate a 50 Hz sinusoidal AC high voltage and a positive- or negative-polarity DC high voltage, and measure the RMS values of AC voltage and current and the amplitude/average values of DC voltage and current, for high-voltage testing and insulation diagnostics of power cables, surge arresters, solid dielectrics, protective equipment and other objects and materials that require high-voltage testing.

The instrument is available in 12 modifications, which differ in the design of the control unit (CU) and high-voltage unit (HVV), the polarity of the DC output, the maximum output/measurement values of DC and AC voltage and current, and the permissible measurement uncertainty limits.

Modification	Configuration
SKAT-70C, SKAT-70CA, SKAT-100C, SKAT-100CA	CU in a portable case; HVU in a steel enclosure
SKAT-70C-K, SKAT-70CA-K	CU in a portable case; HVU in a plastic enclosure
SKAT-70C-H, SKAT-70CA-H	CU in a metal enclosure with carrying handle; HVU in a steel enclosure
SKAT-70C-S, SKAT-70CA-S, SKAT-100C-S, SKAT-100CA-S	CU in a 19" rack-mount enclosure; HVU in a steel enclosure

DC output polarity is negative for the SKAT-70C, SKAT-70C-K, SKAT-70C-S, SKAT-70C-H, SKAT-100C and SKAT-100C-S modifications, and positive for the SKAT-70CA, SKAT-70CA-K, SKAT-70CA-S, SKAT-70CA-H, SKAT-100CA and SKAT-100CA-S modifications.

The instrument is designed for operation at an ambient temperature of $-10\text{ }^{\circ}\text{C}$ to $+40\text{ }^{\circ}\text{C}$ and relative humidity up to 95 % at $25\text{ }^{\circ}\text{C}$.

3 Embedded Software

The instrument runs embedded firmware; its identification data are given in Table 1.

Table 1 — Software identification data

Identification data	Value
Software identifier	SKAT-C-series
Version number (software ID)	1.0 or later
Digital software identifier	—

The firmware is stored in the microcontroller's non-volatile memory and is metrologically significant; the metrological characteristics of the instrument are rated with the influence of the firmware taken into account.

The instrument design precludes unauthorized interference with the firmware or with measurement data. The firmware can be verified, installed or reinstalled only at the manufacturer's facility, using dedicated hardware and software tools. The firmware's protection level against unintentional and deliberate modification, per R 50.2.077-2014, is **"high."**

4 Metrological and Technical Specifications

Table 2 — Metrological characteristics

Characteristic	SKAT-70C family ¹	SKAT-100C family ²
AC output voltage range, 50 Hz sinusoidal, RMS, kV	1 to 51	1 to 71
DC output voltage range, either polarity, kV	1 to 71	1 to 101
AC voltage measurement range, RMS, kV	2 to 50	3 to 70
DC voltage measurement range, either polarity, amplitude, kV	3 to 70	5 to 100
Permissible relative measurement uncertainty, AC RMS voltage, %	$\pm(2.0+0.04 \cdot ((50/U)-1))^*$	$\pm(2.0+0.04 \cdot ((70/U)-1))^*$
Permissible relative measurement uncertainty, DC voltage (amplitude), %	$\pm(2.0+0.04 \cdot ((70/U)-1))^*$	$\pm(2.0+0.04 \cdot ((100/U)-1))^*$
Maximum AC output current, mA	50	35
Maximum DC output current, mA	20	15
AC current measurement range, RMS, mA	0.1 to 50	0.1 to 35
DC current measurement range, either polarity, average, mA	0.1 to 20	0.1 to 15
Permissible relative measurement uncertainty, AC current, %	$\pm(2.0+0.1 \cdot ((50/I)-1))^*$	$\pm(2.0+0.1 \cdot ((35/I)-1))^*$
Permissible relative measurement uncertainty, DC current, %	$\pm(2.0+0.1 \cdot ((20/I)-1))^*$	$\pm(2.0+0.1 \cdot ((15/I)-1))^*$
Test DC voltage ripple factor, %, max.	3	
AC test voltage sinusoidal distortion factor, %, max.	5	

1 SKAT-70C, SKAT-70CA, SKAT-70C-K, SKAT-70CA-K, SKAT-70C-S, SKAT-70CA-S, SKAT-70C-H, SKAT-70CA-H.

2 SKAT-100C, SKAT-100CA, SKAT-100C-S, SKAT-100CA-S. *U – measured voltage, kV; I – measured current, mA.

Table 3 — General technical characteristics

Characteristic	Value
Supply voltage, V	198 to 242
Supply frequency, Hz	49 to 51
Supply voltage distortion factor, %, max.	5
Maximum apparent power consumption, V·A	2800
Maximum AC output apparent power, V·A, not less than	2000

Maximum DC output power, W, not less than	700
Max. continuous AC run time per duty cycle (min. 20 min OFF between cycles) — above 1.8 kW / 1.5–1.8 kW / 1.0–1.5 kW / up to 1.0 kW	3 min / 6 min / 15 min / 3 h
Max. continuous DC run time per duty cycle (min. 20 min OFF between cycles) — above 0.6 kW / 0.4–0.6 kW / up to 0.4 kW	3 min / 10 min / 1 h
Ambient operating temperature, °C	–10 to +40
Relative humidity at +25 °C, %, max.	95
Atmospheric pressure, kPa	84 to 106.7
Mean service life, years, not less than	10
Mean time between failures, h, not less than	8000
Mean time to restore, h, not less than	8

The instrument provides Manual and Automatic test modes. In Automatic mode the instrument raises the voltage at the rate set by the operator to the target test voltage, in steps of 0.1 kV, and holds it stable. In Manual mode the rotary encoder directly drives the voltage regulator. A built-in **Program** test mode is also available: the instrument automatically ramps to the pre-set voltage at the pre-set rate, holds it for the pre-set time, removes the voltage and stores a test report in non-volatile memory. Ten programs can be created and edited from the user menu, and the reports of the last 50 program tests can be reviewed and printed on the built-in printer. A real-time clock is also provided.

A built-in changeover switch selects AC or DC output without the need to manually fit a jumper or diode module inside the high-voltage unit. The same switch discharges the residual capacitive charge through the secondary winding's resistance at the end of a test.

Output voltage is measured directly at the load, which improves both measurement accuracy and personnel safety — particularly while a residual capacitive charge is being removed after the high voltage is switched off. Voltage measurement and regulation accuracy fully comply with GOST 1516.2-97 for insulation test equipment used on power cables and solid dielectrics. The recalibration interval is 2 years.

The output current is hardware-limited to the set value, and the high-voltage transformer is protected against overheating. On a load breakdown, the instrument disconnects the high voltage and captures the breakdown voltage on the display. An automatic grounding rod is provided for additional safety, together with a connector for an external high-voltage interlock and an external signal lamp.

An automatic **Burn** (pre-conditioning) mode is also provided, used to reduce a defective joint's or a "self-healing" fault's breakdown voltage down to the burn-in equipment's operating level, and it can also be used to charge the capacitors of an acoustic shock-wave (fault-locating) generator.

5 Design and Construction

The instrument consists of two units — a high-voltage unit (HVU) and a control unit (CU) — connected by an interconnecting cable. The external appearance of the modifications is shown in Figures 1–6.



Control unit



High-voltage unit

Figure 1. SKAT-70C / SKAT-70CA



Control unit



HV unit, plastic enclosure

Figure 2. SKAT-70C-K / SKAT-70CA-K



Figure 3. SKAT-70C-H / SKAT-70CA-H (metal enclosure with handle)



Figure 4. SKAT-70C-S / SKAT-70CA-S (19" rack-mount)



HV unit, mobile cart



Control unit

Figure 5. SKAT-100C / SKAT-100CA



Figure 6. SKAT-100C-S / SKAT-100CA-S (19" rack-mount)

The instrument's operating principle is based on converting the single-phase AC mains supply into a regulated AC high voltage by means of a voltage regulator and a high-voltage transformer. In AC mode, the high voltage is routed to the output through a high-voltage changeover switch. In DC mode, the regulated AC high voltage is converted to rectified DC by a half-wave rectifier before reaching the output.

Output voltage is measured with a high-voltage divider connected directly at the HV unit's output, so the voltage actually present at the instrument's output is always what is measured. Output voltage and current readings are shown on the control unit's display.

5.1 Control Panel



Figure 7. Control unit front panel

Control / Connector	Function
Display	Liquid-crystal display; shows the mode, all test parameters and progress, and the measured voltage and current.
HIGH VOLTAGE indicator	Lit while the output voltage is energized.
START / STOP buttons	Start and stop the test.
Encoder knob ("CONTROL")	Adjusts test parameters and, in Manual mode, drives the output voltage directly; rotate to change a value / select a menu item, press to confirm.
POWER button	Switches the instrument's mains power on.
TRANSFORMER connector	Connects the interconnecting cable to the high-voltage unit.
Grounding terminal	Instrument protective-earth connection point. The unit must be grounded at all times.
USB port	Firmware updates and (on some models) report/sequence export.
RS-232 port	Optional connection for an external printer (not fitted on rack-mount and handle-case modifications, which have a built-in thermal printer instead).

INTERLOCK connector	Optional connection for an external HV-on indicator (audible or visual) and an external high-voltage interlock circuit — see Figure 8. Dry contacts close when the high voltage is on; max. 1 A, 250 V AC / 30 V DC. Leave unconnected if not required.
Mains inlet & 15 A fuses	Mains supply connection and overcurrent protection.
EMERGENCY OFF button	Immediately removes the high voltage.

Figure 8 — Interlock and external signaling wiring

Pin	Circuit
1, 2	Door interlock (S1)
3, 4	External lamp (H1)
5, 6	Connector-present sensor

S1 – door sensor; H1 – external signal lamp; A1 – instrument control unit; XP1 – control unit INTERLOCK connector.

On the SKAT-100C-S, SKAT-100CA-S, SKAT-70C-S, SKAT-70CA-S, SKAT-70C-H and SKAT-70CA-H modifications, the mains inlet, TRANSFORMER and INTERLOCK connectors and the grounding terminal are located on the rear panel; these modifications have a built-in thermal printer on the front panel and no RS-232 port.

The high-voltage unit is fitted with a grounding rod, operated from the control unit, which grounds the test object after its stored charge has been removed (capacitive loads); the rod retracts from the HV terminal while a measurement is in progress.

Unit nameplates (metallic or polymer self-adhesive labels) state the product name, date of manufacture and serial number, and are fixed to the outer surface of the control unit's case lid and to the side of the high-voltage unit.

The instrument is packed in an individual plywood crate; the control unit and high-voltage unit are secured inside the crate with wooden battens to prevent movement. Accessories (see Section 10, Scope of Delivery) are packed in plastic bags, with the documentation in a separate plastic bag.

6 Safety Instructions

Observe general high-voltage workplace safety rules whenever operating the instrument.

DANGER

Electric Shock Hazard

Never assume equipment is safe to handle without using the required safety equipment and grounding procedures.

- All procedures must comply with local safety regulations.
- Always treat exposed connectors and conductors as potential electric-shock hazards.
- The device under test (DUT) must be grounded, de-energized and isolated from all power sources before connection.
- All cables and connectors must be inspected for damage before use; damaged equipment must not be used.
- Ground connections must be made first and removed last.
- The DUT must be discharged and grounded before disconnecting the test lead.

DANGER

Untrained / Unqualified Personnel

Operating an ungrounded instrument is prohibited. Do not connect the control unit and high-voltage unit to a grounding bus in series (i.e., daisy-chained) — each must be grounded independently. Only qualified electrical personnel who have studied this manual and mastered safe working methods for this equipment, hold electrical safety qualification group III or higher, and are authorized for independent work in installations above 1000 V, may operate the instrument. Personnel who have not passed the required assessment must not be permitted to work with the instrument.

The instrument casing, controls, measuring leads and accessories must be free of mechanical damage. The grounding terminal must be clearly marked. Grounding contact surfaces must be undamaged, clean, smooth and free of oxidation or corrosion. All connections must be securely made and undamaged. The mains plug's grounding contacts must be in good condition and provide reliable electrical contact.

All personnel operating or servicing the instrument must be familiar, to the appropriate extent, with the Russian Federation's electrical installation operating rules (PTEEP, Order of the Ministry of Energy No. 6 of 13.01.2003) and occupational safety rules for electrical installations (POTEU, Order of the Ministry of Labor No. 903N of 15.12.2020), or the equivalent local regulations in the country of use. The workplace must comply with fire-safety requirements (GOST 12.1.004-76 or local equivalent).

7 Preparation for Operation

Before testing, connect the control unit and the high-voltage unit using the interconnecting cable. Both units must be grounded.

DANGER

Grounding Required

Operating the instrument without grounding is prohibited. Do not connect the control unit and the high-voltage unit to the grounding bus in series — each unit must have its own independent connection to ground.

The test object must be grounded at one end and connected to the instrument's high-voltage output at the other. On the SKAT-70C-K and SKAT-70CA-K modifications, the high-voltage unit's lid must be open, with the lead connected to the HV output routed to the right (away from the grounding rod).

Verify that the power button responds crisply by switching the control unit off and back on. On power-up, the display shows the modification and firmware series, then the measured output voltage and current at zero. Pressing **START** sounds a tone, retracts the HV unit's grounding rod, and lights the **HIGH VOLTAGE** indicator on the control unit. Pressing **STOP** extinguishes the **HIGH VOLTAGE** indicator and returns the grounding rod to ground the instrument's HV output.

If the interlock and external signaling circuits are used, verify that they function correctly. The high-voltage unit must be kept clear of any metal structure by a distance sufficient to prevent flashover.

NOTICE

DC Testing of Non-Capacitive Loads

When testing in DC mode on a load that is not itself capacitive, connect an additional high-voltage capacitor in parallel with the test object. Its capacitance must be at least 0.01 μF and must satisfy:

$$C > 1 / (3 \cdot R), \mu\text{F}, \text{ where } R \text{ is the load resistance in } \text{M}\Omega.$$

If this condition is not met, the test DC voltage ripple will exceed 3 %. Recommended capacitor type: IK-100-0.25. A ballast capacitor is not required when testing capacitive loads such as cable lines.

8 Operating Instructions

8.1 Selecting a Mode

The operating mode is selected from the main menu. To open the main menu, press and hold the **CONTROL** encoder for one second while the high voltage is off. Rotate the encoder to highlight the desired mode and press briefly to select it. Modes marked **AC** operate on alternating current, those marked **DC** on direct current.

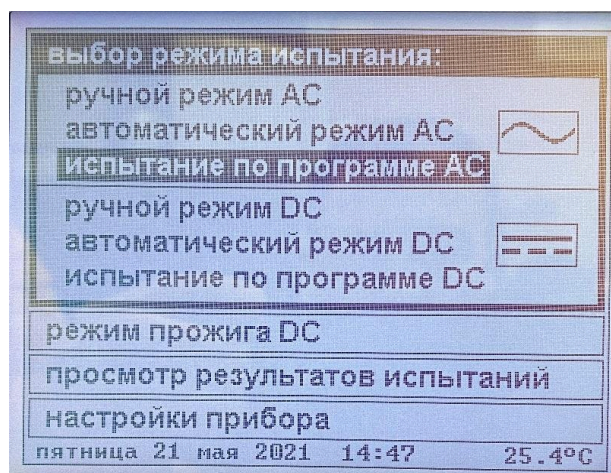


Figure 9. Main menu screen

8.2 On-Screen Information

During a test the display shows: output type — AC or DC (top line); the parameters of the selected mode (upper section); the measured output voltage and current (lower section); auxiliary readings (middle-left); the test time (middle-right); and the date, time and internal HV-unit temperature (bottom line).



Figure 10. Manual mode screen

In AC tests, the lower section shows the RMS output voltage and current, while the auxiliary section also shows the peak output voltage. In DC tests, the lower section shows the peak output voltage and the average current, while the auxiliary section shows the RMS output voltage. The auxiliary section also shows the instrument's output power and the average current over the last 10 seconds (leakage current).

8.3 Manual Mode

Manual-mode parameters can only be changed while the high voltage is off. To change **Max. Voltage** or **Max. Current**: rotate the encoder to highlight the parameter, press briefly, rotate to set the value, and press briefly again to confirm. **Timer**, **On Timeout** and **Timer Start** are changed the same way, by highlighting and briefly pressing the encoder.

If **Timer** is set to *count up*, timing starts from zero; the start is governed by **Timer Start**: *automatic* starts timing when the high voltage is switched on, *manual* starts it on a second press of **START** during the test. If **Timer** is set to *countdown*, a second press of **START** during the test begins the countdown from the pre-set value. To set the countdown time, highlight the time field, press the encoder briefly, set the minutes, press again, set the hours, and press a third time. When the countdown reaches zero, the test ends automatically (if **On Timeout** is set to *end test*) or an audible signal sounds, which can be silenced by pressing **START**.

Press **START** to begin the test. The instrument automatically retracts the grounding rod, sets the internal HV changeover switch to the selected output type, and energizes the regulator's primary winding (the **HIGH VOLTAGE** indicator lights). During the test the active mode is highlighted on screen, and rotating the **CONTROL** encoder drives the voltage regulator's stepper motor; the output is limited as it approaches the **Max. Voltage** setting.

WARNING

In DC Manual mode, change the output voltage gradually. After you stop turning the encoder, the output will continue to rise slightly as the load capacitance finishes charging.

Press **STOP** to end the test. The regulator returns to zero, the primary winding is de-energized, the **HIGH VOLTAGE** indicator goes out, the capacitive test object is discharged (DC tests), and the grounding rod re-engages.

8.4 Automatic Mode

Automatic-mode parameters can only be changed while the high voltage is off. **Max. Current**, **Timer**, **On Timeout** and **Timer Start** (and the countdown time) are set the same way as in Manual mode.



Figure 11. Automatic mode screen

To set the voltage ramp rate, highlight **Rate** with the encoder, press briefly, rotate to the desired value and press again to confirm. The **VOLTAGE** parameter sets the target test voltage; while the high voltage is off it is set the same way. During the test, turning the encoder changes this setpoint and the instrument re-stabilizes the output at the new value.

Press **START** to begin. The instrument retracts the grounding rod, sets the HV changeover switch, and energizes the primary winding (the **HIGH VOLTAGE** indicator lights), then automatically ramps to the target voltage at the set rate. If **Timer** is *count up* and **Timer Start** is *manual*, timing starts on a second press of **START**; in all other cases the timer starts automatically once the output reaches the **VOLTAGE** setpoint.

The test ends when **STOP** is pressed, or automatically at the end of the countdown (if **On Timeout** is set to *end test*). The regulator returns to zero, the primary winding is de-energized, the **HIGH VOLTAGE** indicator goes out, the capacitive test object is discharged (DC tests), and the grounding rod re-engages.

8.5 Program Test Mode

In Program mode, briefly press the **CONTROL** encoder while the high voltage is off to step through the stored programs.

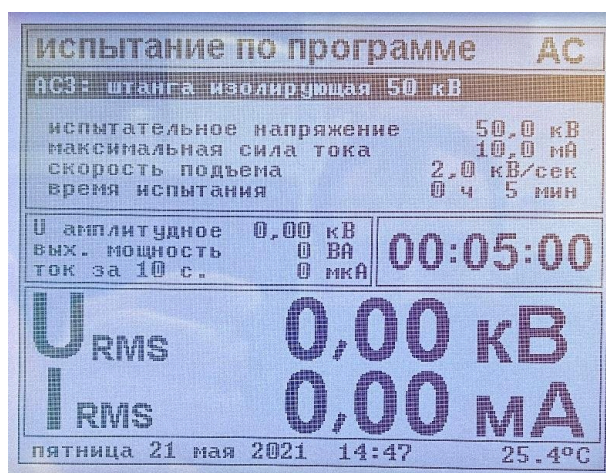


Figure 12. Program test mode screen

Press **START** to run the selected program. The instrument retracts the grounding rod, sets the HV changeover switch, energizes the primary winding, and automatically runs the test with the program's pre-set parameters. On completion, a detailed report is saved to the instrument's internal memory. Press **STOP** at any time to abort the test.

8.6 Burn (Pre-Conditioning) Mode

With the high voltage off, set the burn time and maximum voltage by highlighting each field with the encoder, pressing briefly, rotating to the required value, and pressing again. The maximum voltage in Burn mode is 60 kV for the SKAT-70C, SKAT-70CA, SKAT-70C-K, SKAT-70CA-K, SKAT-70C-S, SKAT-70CA-S, SKAT-70C-H and SKAT-70CA-H modifications, and 80 kV for the SKAT-100C, SKAT-100CA, SKAT-100C-S and SKAT-100CA-S modifications. Maximum burn time is 3 minutes; allow at least a 5-minute pause between burn cycles.



Figure 13. Burn mode screen

Burn mode operates automatically once started with **START**. The instrument ramps up the voltage; on breakdown of the load, a countdown of the burn time begins, and the instrument delivers the maximum available current throughout. Press **STOP** to abort manually, or the process ends automatically when the countdown reaches zero — in both cases the regulator returns to zero, the primary winding is de-energized, the **HIGH VOLTAGE** indicator goes out, the test object is discharged, and the grounding rod re-engages.

8.7 Editing Test Programs

From the main menu, use the **CONTROL** encoder to open **Instrument Settings**, then select the program to edit.

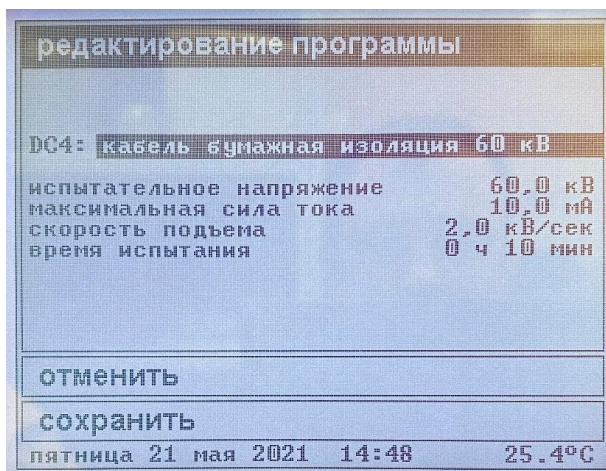


Figure 14. Program editing screen

The instrument stores five AC test programs (AC1–AC5) and five DC test programs (DC1–DC5). For each program you can set its name, test voltage, maximum current, ramp rate and test time. Highlight the field to edit with the encoder, press to confirm, rotate to set the value, and press again.

To edit a program's name, highlight it and press the encoder to enter text-entry mode.

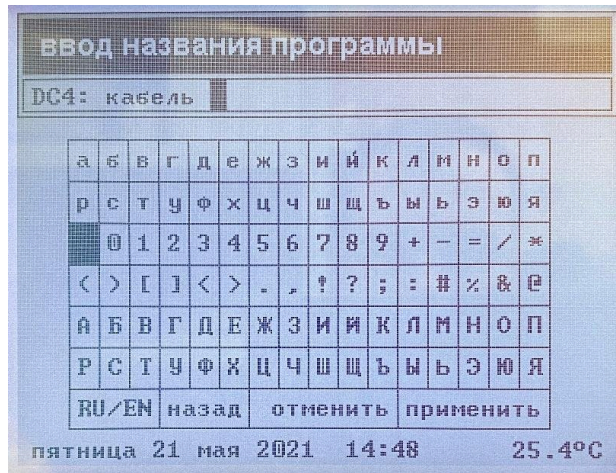


Figure 15. Program name entry screen

RU/EN switches the on-screen character set between Cyrillic and Latin layouts; **back** deletes the last character entered (this can also be done by pressing **STOP** with any character/button highlighted). Select **apply** to accept the new name, or **cancel** to discard it. To save the program, select **save** in the program editing window; selecting **cancel** discards the changes. The factory-default programs can be restored from **Instrument Settings** → **Restore Factory Programs**, confirmed by pressing **START**.

8.7.1 Setting the Date and Time

From the main menu, open **Instrument Settings** with the encoder, then select **Set Date and Time**.

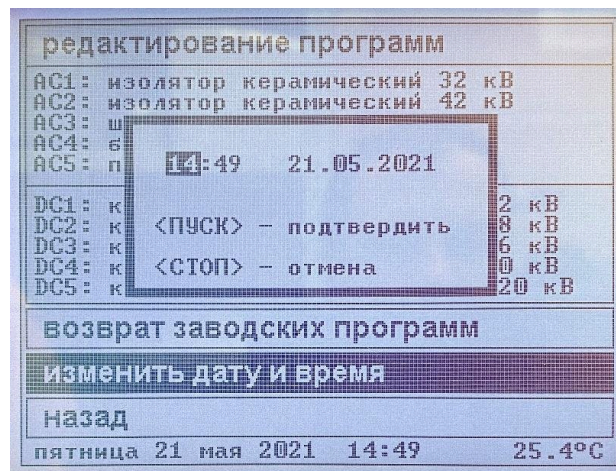


Figure 16. Date/time setting screen

While setting the clock, rotating the encoder changes the highlighted value and pressing it cycles the highlight through minutes, hours, year, month and day. Press **START** to confirm, or **STOP** to cancel.

8.8 Viewing and Printing Test Results

The instrument stores the results of the last 50 program tests. To view them, open **View Test Results** from the main menu and select the desired report; reports are listed in order, with the most recent as No. 1 and the oldest at the end of the list. Each report records the test date and time, the program used, the test parameters and the outcome. For a successful test, the current is logged at 11 points: at the start of the test (once the target voltage is reached) and at 10 %, 20 %, 30 % ... 100 % of the test time. If a breakdown occurs, the voltage and current at the moment of breakdown are recorded; a report is likewise saved on overload or overheating.

Press the **CONTROL** encoder to return to the report list. Press **START** while viewing a report to print it.
Press **STOP** to exit the results view.

8.9 Service Messages

The instrument has a built-in self-diagnostic system. On a fault, an error message is displayed (e.g. "HV Unit Connection Error," "Error: Regulator Not Returning to Zero," "Error: Regulator Not Connected," "Transformer Fault," "Changeover Switch Fault or Diode Breakdown"). If errors recur systematically, contact the manufacturer.

Messages are also shown on a load breakdown ("Load Breakdown Detected"), on high-voltage unit overheating ("Transformer Overheating"), and when the external door-interlock sensor trips the high-voltage cutoff ("WARNING! Door Open"). Press **STOP** to clear a service message.

9 Storage and Transportation

Packaged instruments may be transported by any mode of transport that protects them from damage, in accordance with the carriage rules applicable to that mode.

Transport conditions with respect to climatic factors must correspond to storage category 5 of GOST 15150 (or the equivalent local standard); with respect to mechanical factors, to group 3 of GOST 22261. Storage conditions with respect to climatic factors correspond to category 2 of GOST 15150.

Secure the packaged instrument on the transport vehicle so that it cannot shift in transit.

10 Scope of Delivery

Table 4 — Scope of delivery

Item	Qty	Remarks
Control unit (per ordered modification)	1	—
High-voltage unit (per ordered modification)	1	—
Interconnecting cable	1	(4±0.1) m
Mains cable	1	(4±0.1) m, 250 V, 16 A
Grounding cable	2	(4±0.1) m, 4 mm ²
Fuse, 15 A	2	Ceramic, 6.35×30 mm
Signal-circuit connector plug	1	FQ14-6TJ
Technical Passport	1	STSK.411728.014-2022 PS
User Manual	1	STSK.411728.014-2022 UM

11 Disposal

When the instrument reaches the end of its service life and repair is no longer practicable or economically justified, it must be decommissioned and disposed of as follows:

- Drain the transformer oil from the high-voltage transformer's tank into a dedicated container and dispose of it in accordance with the applicable requirements for used oil and lubricants (e.g. GOST R 52108-2003 or the equivalent local standard).
- Disassemble the instrument into its component parts and dispose of them in accordance with the applicable standard (e.g. GOST R 52108-2003 or the equivalent local standard).

12 Maintenance and Service

Periodically wipe the high-voltage unit's insulator with ethyl alcohol.

In the event of a fault, the instrument (or the affected assembly) must be repaired at the manufacturer's facility.

Manufacturer: SKAT Industrial Equipment Plant LLC (SKAT Plant LLC)

Address: 37 Poddubnogo St., Volgograd 400040, Russian Federation

Phone: +7 (8442) 26-99-94

E-mail: st@skat-v.com

Technical support and documentation: www.skat-v.com